DECORATED MOSAICS

Facets, bands, isometric 3D, quilts, jigsaws, weaves, beadwork, mazes and spirals

In these mosaics the basic shape is more than a polygon. Each tile of a tiling can be cut into facets lit from one side, so that it stands out like a pyramid, or into concentric bands that shrink towards its centre or one of its corners. Seen in isometric view, cubes and bars turn the plane into stacks, tripods and frameworks, with three shades for the three directions of their faces. Quilts, jigsaw puzzles, woven strips and beadwork, mazes, meanders and whirling spirals complete the set. Colours come from the RYB colour schemes; the ramp and the sweep give the smooth steps that bands and gradients need.

Colour schemes

RYB schemes take hues at fixed steps of the 12-hue wheel from a base hue (here red); when a tiling needs more colours than a scheme has, tints and shades of the same hues are added. The named palettes copy traditional glazed tiles.


monochromatic (red)


analogous (red)


complementary (red)


split complementary (red)


triad (red)


tetrad (rectangle) (red)


square (red)


rainbow (red)


ramp, dark to light (red)


sweep of hues (red)


zellij palette


azulejo palette


girih palette


earth tones

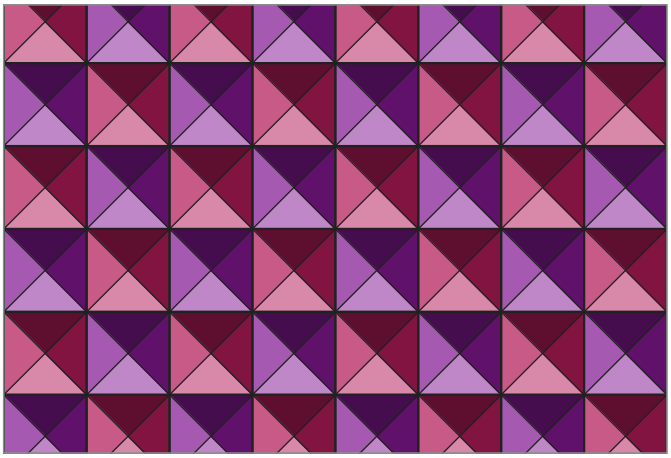

greys

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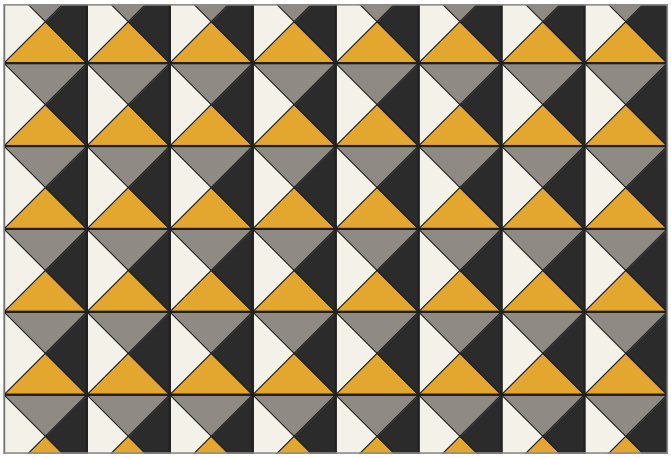
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Facets and pyramids

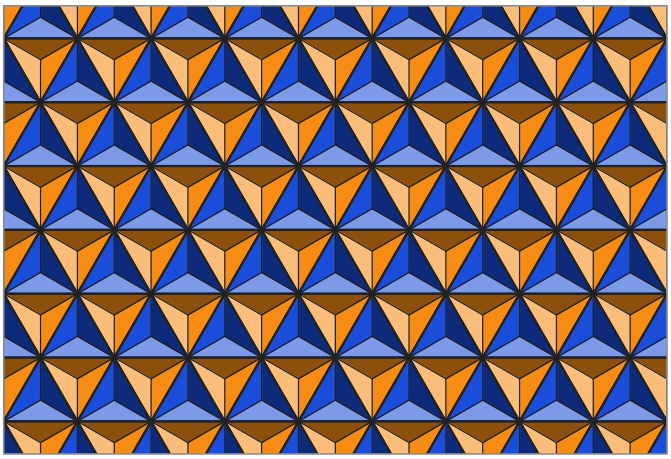
Every tile is cut into triangles from its centre; each triangle is lit according to the direction it faces, so squares become studs, triangles three-dimensional stars and hexagons hexagonal pyramids. Without light the same cut makes the hourglass blocks of quilts. Works on any tiling (-p base=...).



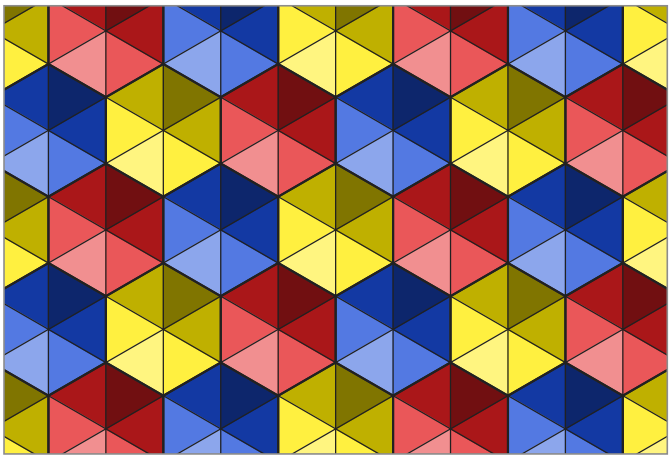
Pyramid studs on a square grid
Colouring: checkerboard · analogous (red-violet)



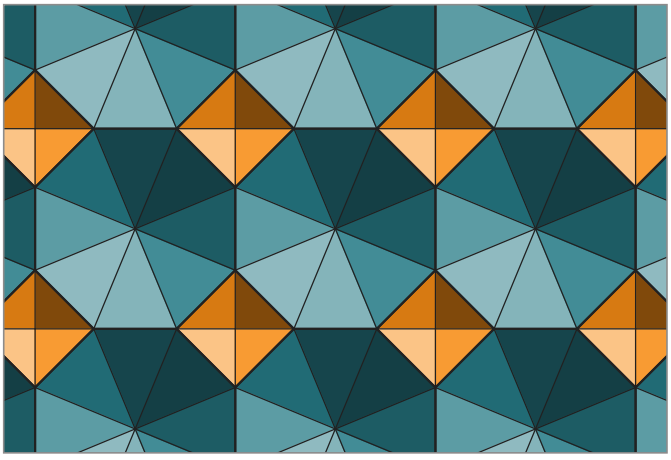
Hourglass blocks: the same cut without light
Colouring: by facet direction



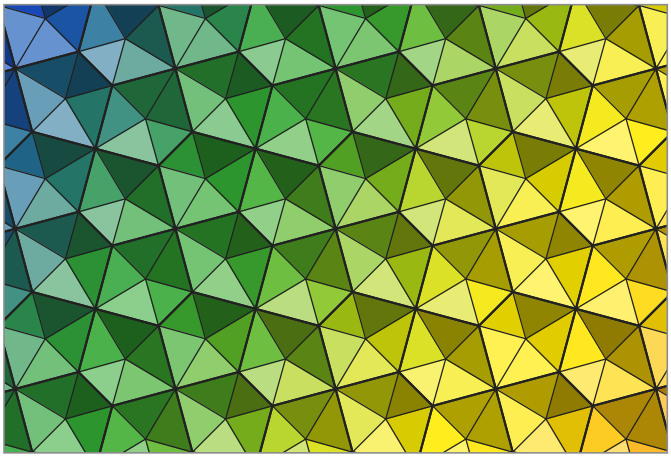
Three-dimensional stars from triangles
Colouring: by tile · complementary (blue)



Hexagonal pyramids in three colours
Colouring: three colours · triad (yellow)



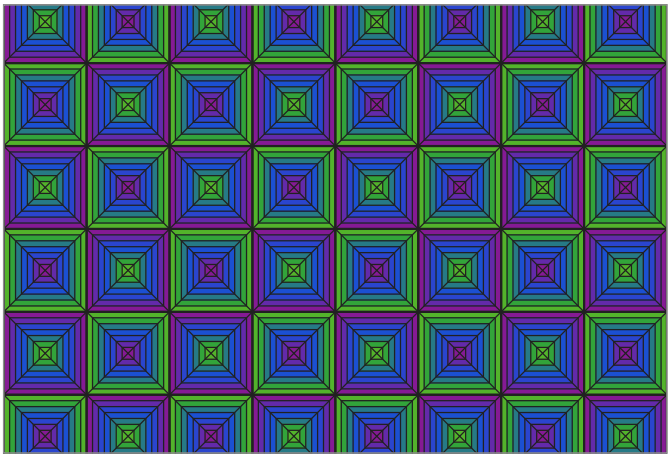
Facets on the 4.8.8 tiling
Colouring: by tile · split complementary (blue-green)



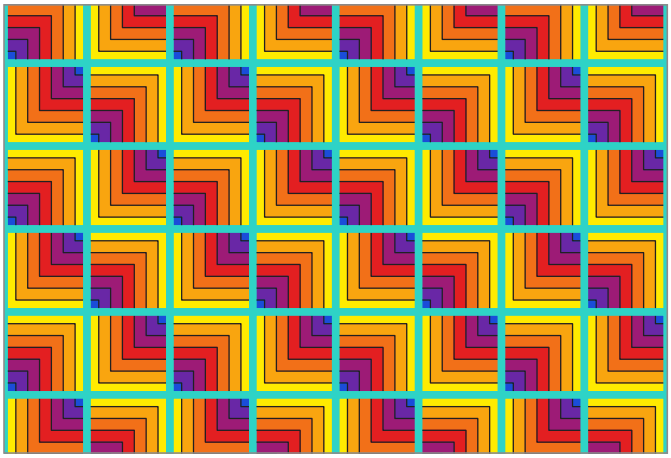
Facets on the Cairo tiling, with a gradient
Colouring: gradient across · sweep of hues (blue)

Concentric bands

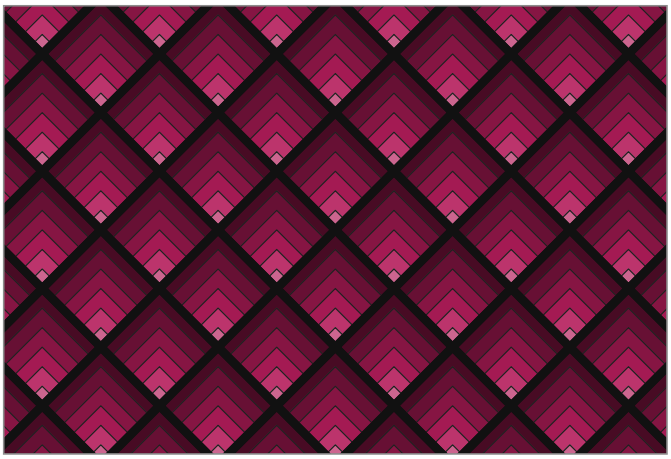
Each tile is filled with bands that shrink towards its centre (courthouse steps) or towards one of its corners (log cabin). Coloured in order with a ramp or a sweep of hues, the bands give depth; a wide gap of another colour frames the blocks.



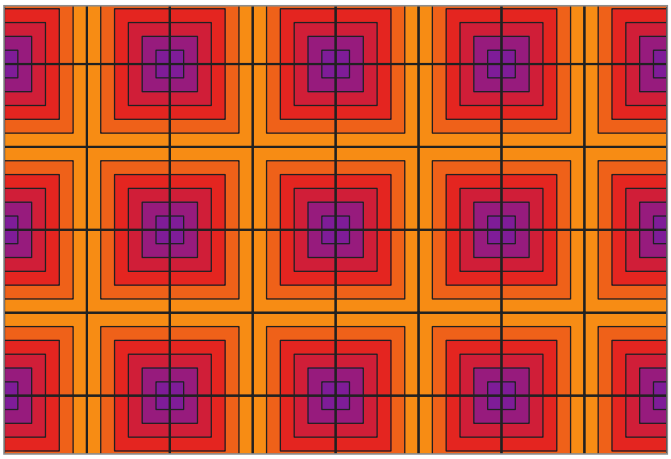
Courthouse steps: rings cut by the diagonals
Colouring: by band, alternating · sweep of hues (violet)



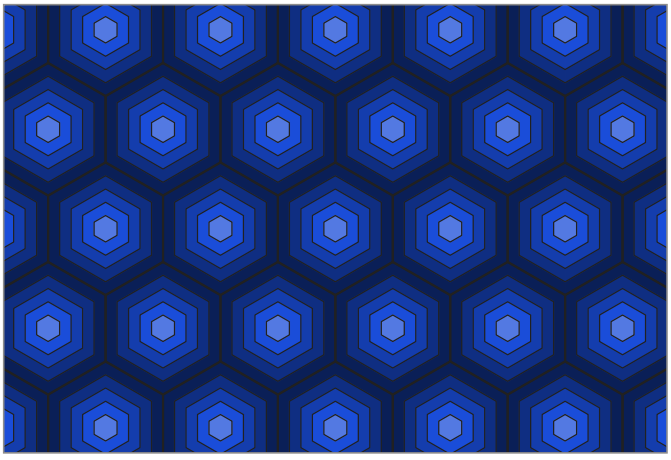
Log cabin: bands from a corner, framed
Colouring: by band · rainbow (yellow)



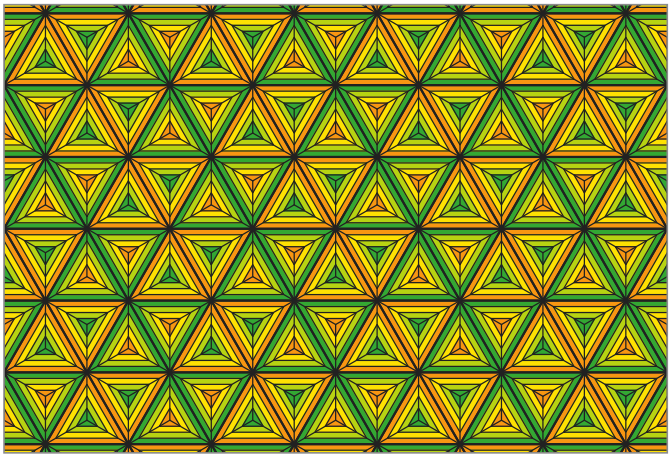
Log cabin diamonds, ramp of one hue
Colouring: by band · ramp, dark to light (red-violet)



Bands round four different corners
Colouring: by band · sweep of hues (orange)



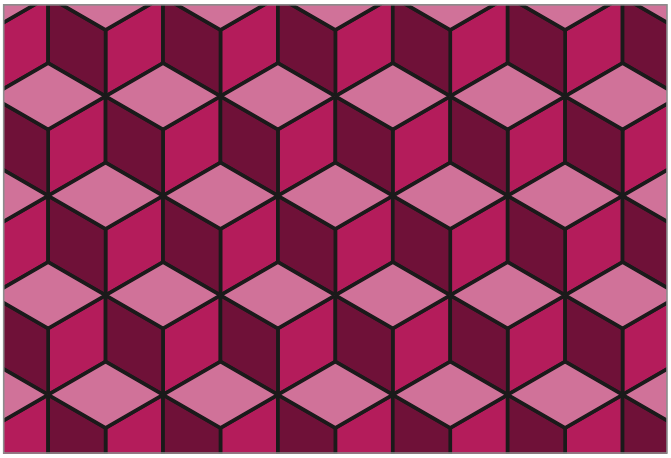
Rings in hexagons
Colouring: by band · ramp, dark to light (blue)



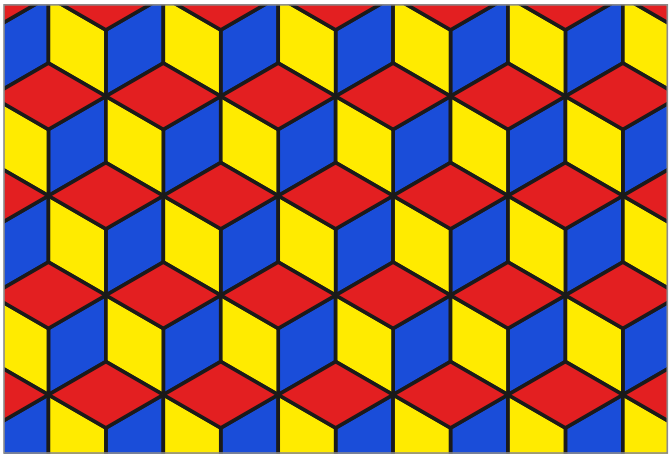
Striped triangles, alternating
Colouring: by band, alternating · sweep of hues (green)

Isometric cubes

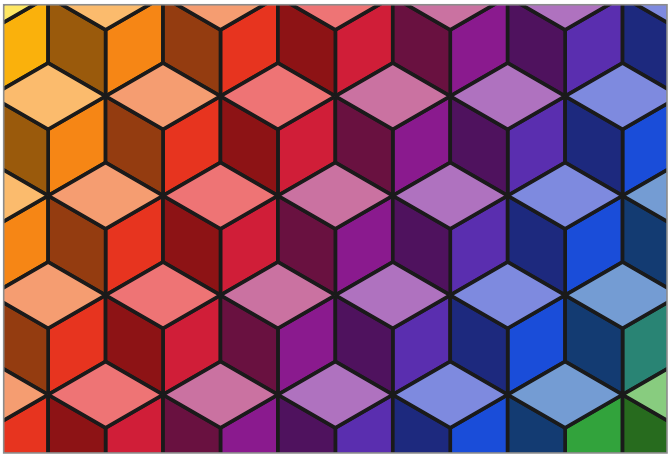
Cubes seen along their diagonal: three faces, one light, one mid, one dark. Stacked on a plane they make the tumbling blocks pattern; cut into stickers they become Rubik's cubes, here with a rainbow that spreads from the centre.



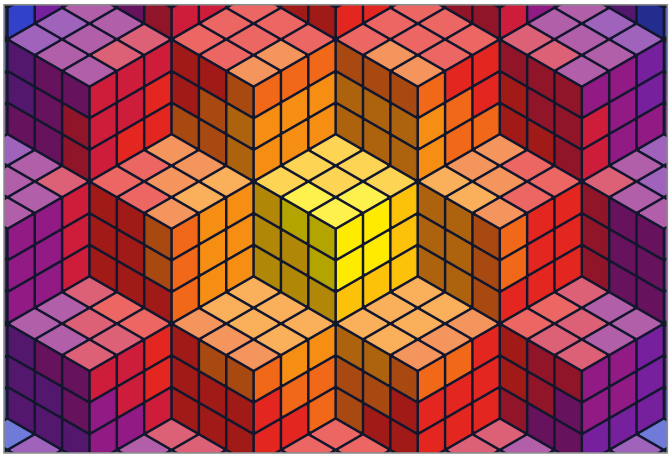
Tumbling blocks: one hue in three values
Colouring: one colour · monochromatic (red-violet)



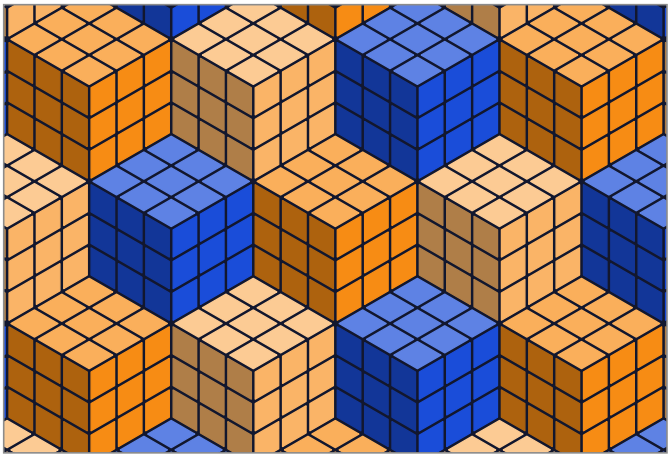
One colour for each direction of face
Colouring: by face · triad (red)



Cubes in a rainbow gradient
Colouring: gradient across · rainbow (yellow)



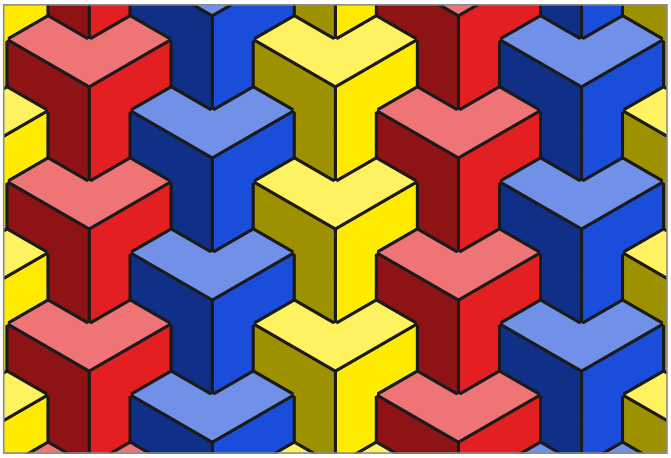
Rubik's cubes, rainbow from the centre
Colouring: rings from the centre · rainbow (yellow)



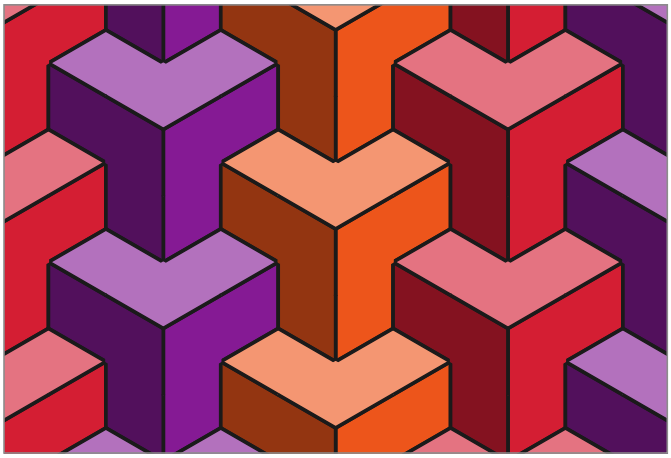
Rubik's cubes in two colours and a tint
Colouring: three colours · complementary (orange)

Tripods and frameworks

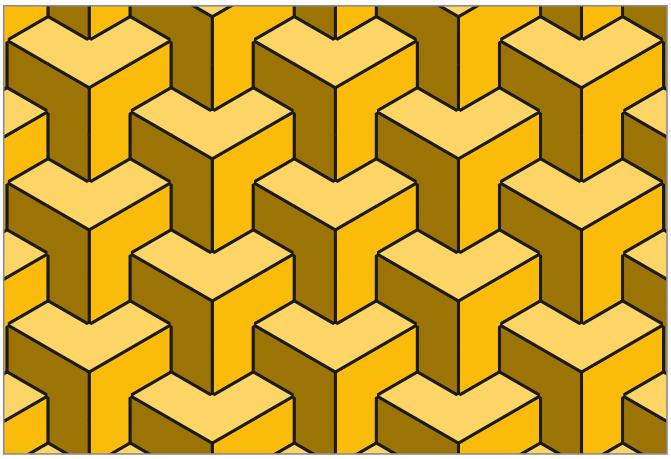
Bars of cubes drawn in isometric view. Three bars meeting at a corner make a Y-shaped tripod, and tripods on the right lattice interlock and cover the plane. Bars along the three axes make a framework with the background showing through.



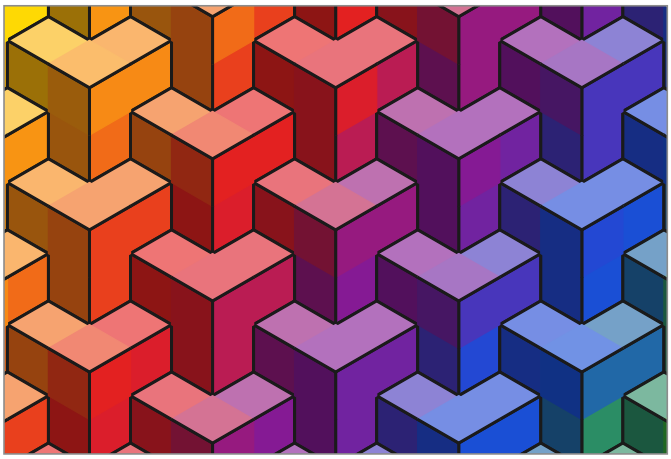
Interlocking Y tripods
Colouring: by object · triad (yellow)



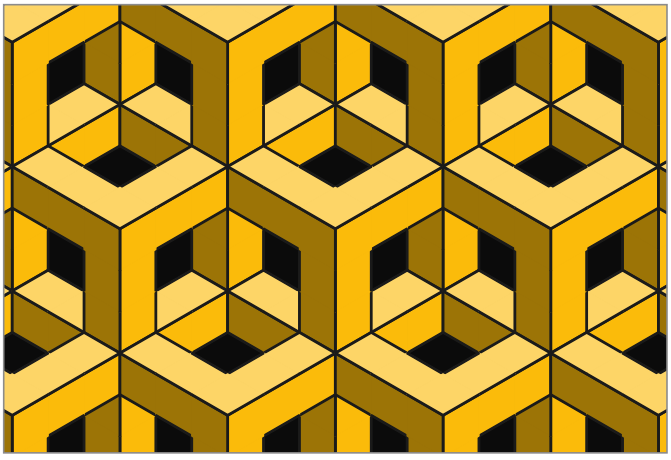
Short tripods in a sweep of warm hues
Colouring: by object · sweep of hues (red-orange)



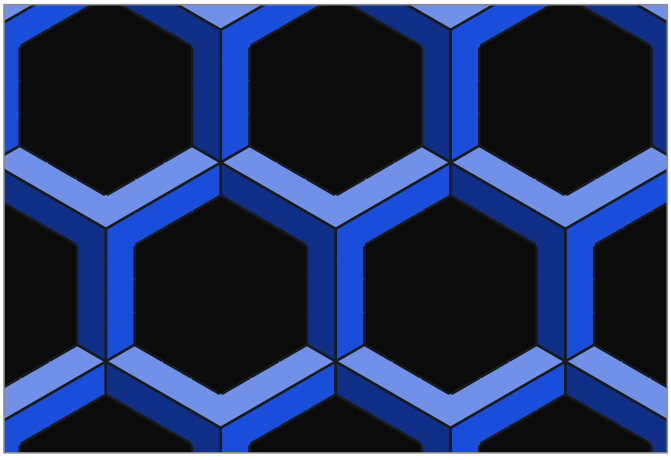
Golden tripods
Colouring: one colour · monochromatic (yellow-orange)



Tripods in a rainbow gradient
Colouring: gradient across · rainbow (yellow)



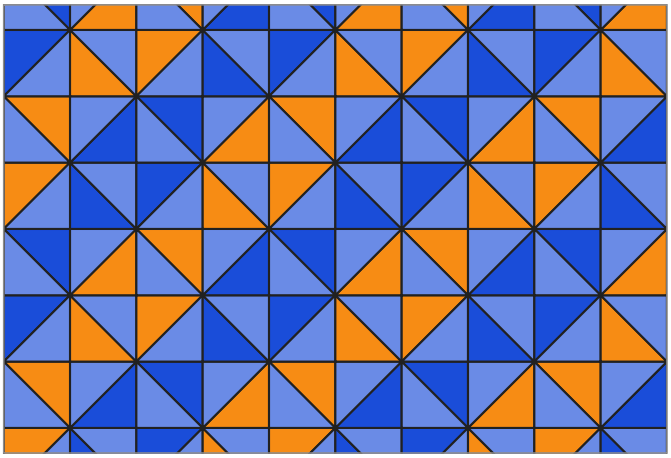
Framework of bars
Colouring: one colour · monochromatic (yellow-orange)



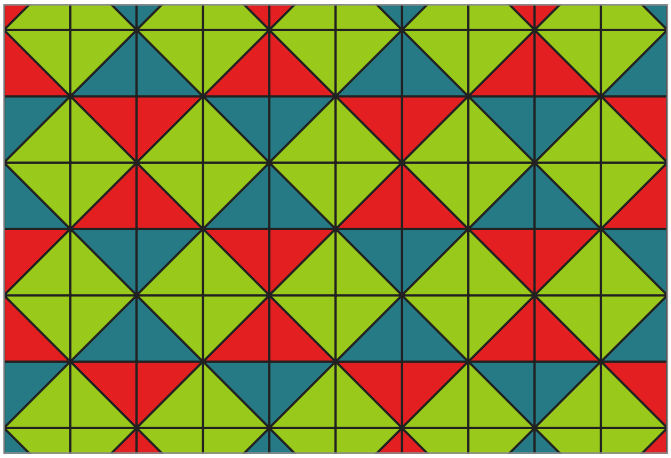
A thinner framework
Colouring: one colour · monochromatic (blue)

Quilts

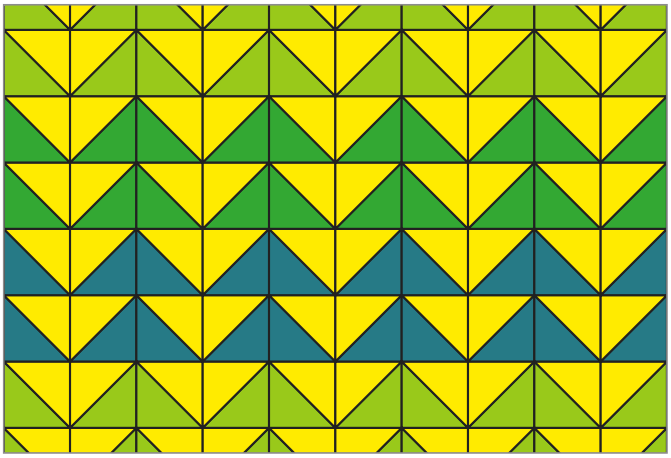
Squares cut along a diagonal into two triangles (half-square triangles), turned according to a small table: pinwheels, diamonds, chevrons, storms. Cut along both diagonals they make the four-triangle hourglass blocks.



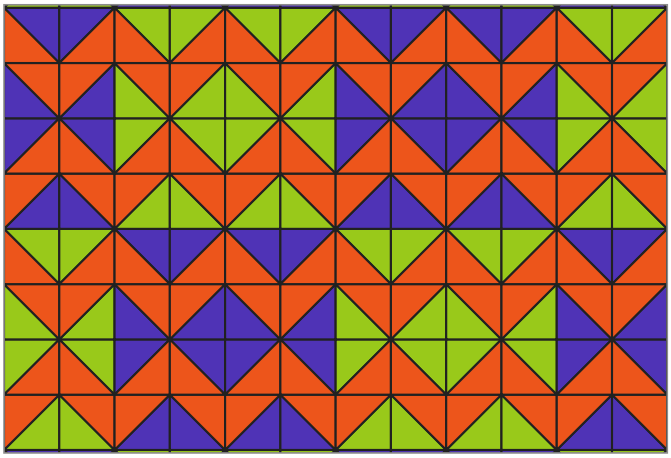
Pinwheels
Colouring: coloured triangles · complementary (blue)



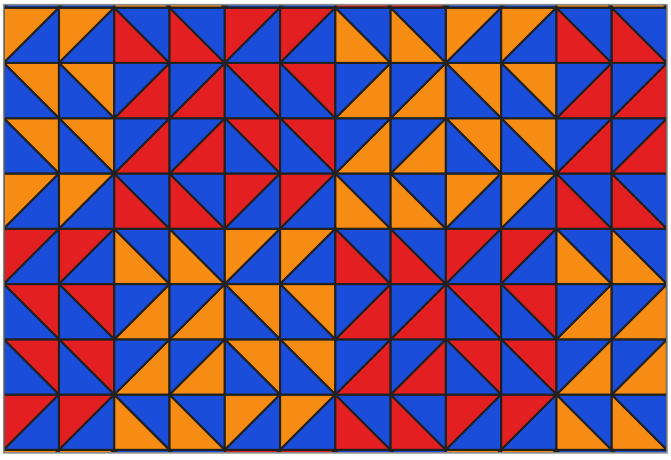
Diamonds
Colouring: coloured triangles · split complementary (red)



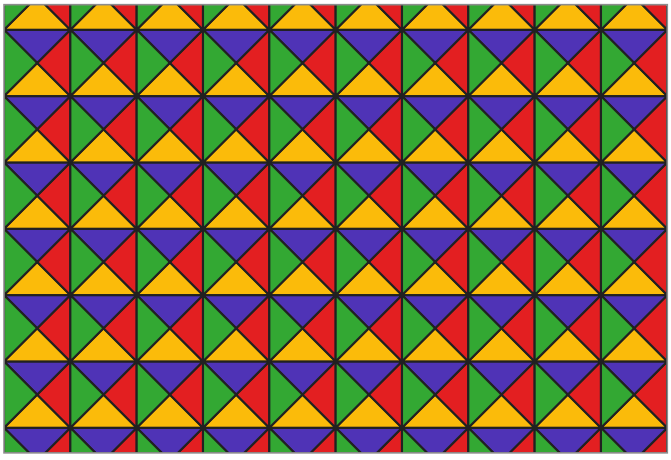
Chevrons
Colouring: coloured triangles · analogous (green)



Storm at sea
Colouring: coloured triangles · triad (blue-violet)



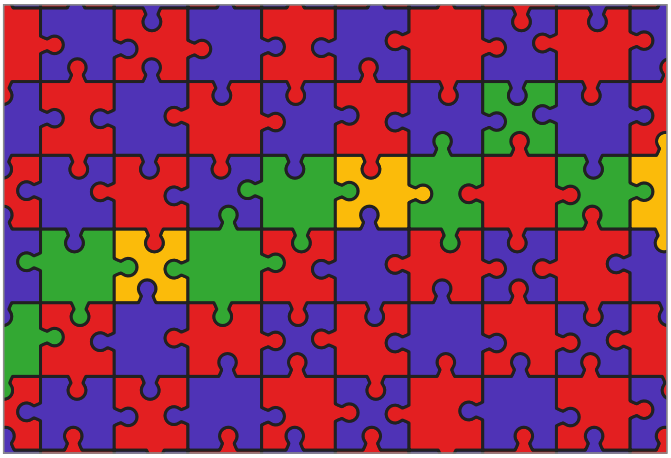
Big pinwheels
Colouring: coloured triangles · tetrad (rectangle) (orange)



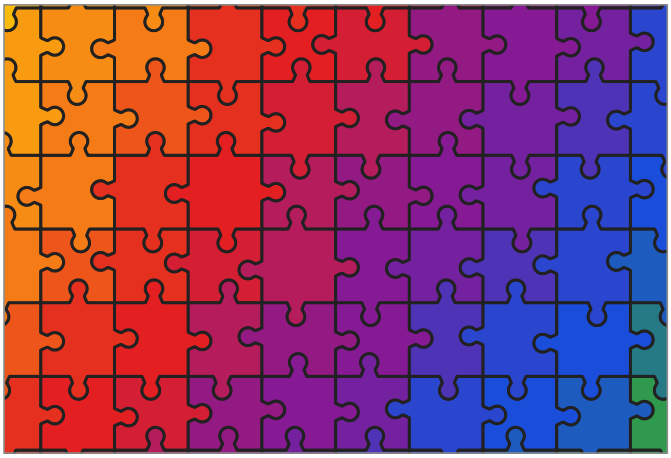
Hourglass: four triangles per square
Colouring: by quarter · square (yellow-orange)

Jigsaw puzzle

Square pieces whose edges carry a tab or a hole, chosen at random for every edge and shared by the two pieces, so that they fit. Colours so that neighbours differ, or a gradient.



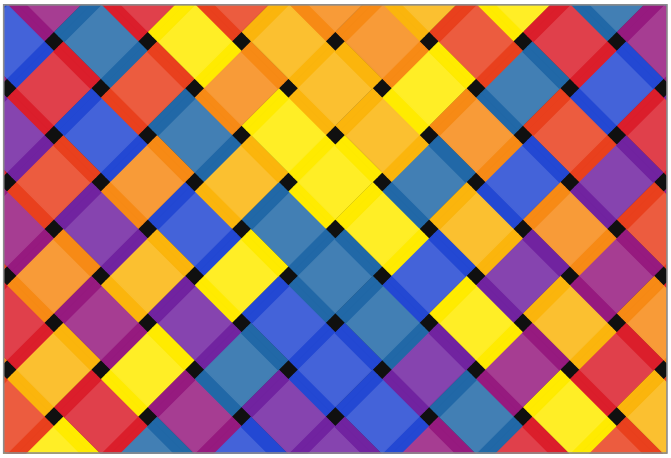
Jigsaw, neighbours differ
Colouring: neighbours differ · square (red)



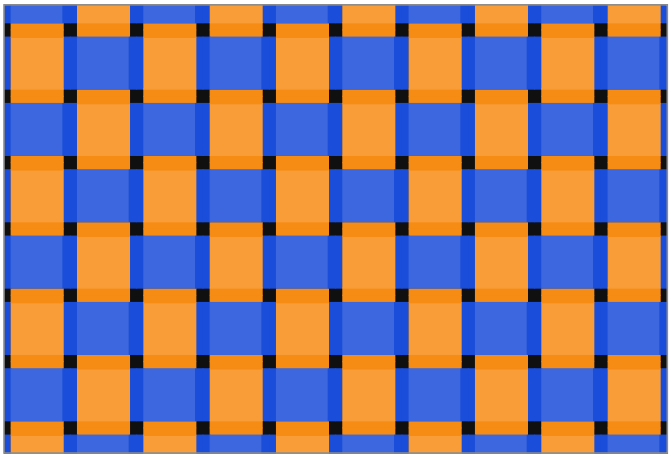
Jigsaw in a gradient
Colouring: gradient across · rainbow (yellow)

Weaves

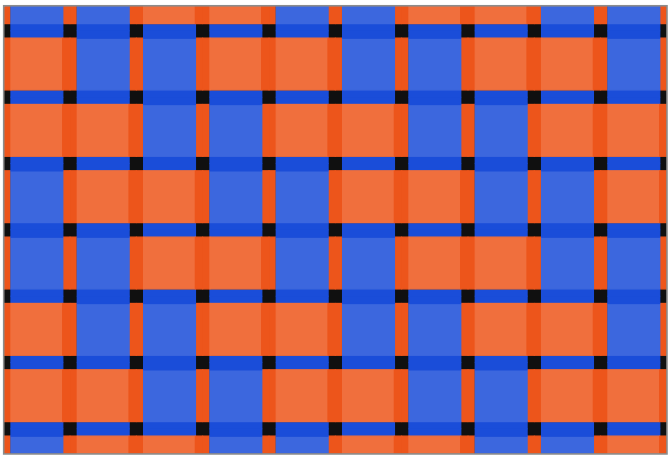
Two families of strips going over and under each other: a plain weave alternates at every crossing, a twill every two, stepping by one each row. A strip darkens where it dives under the other.



Plain weave turned 45°, rainbow strips
Colouring: by strip · rainbow (yellow)



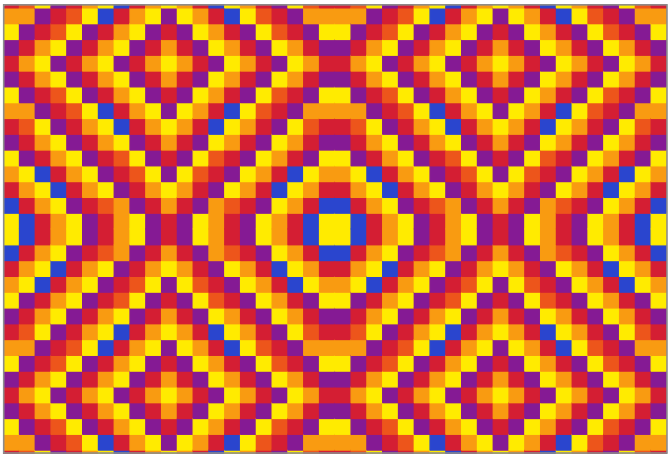
Plain weave in two colours
Colouring: by direction · complementary (blue)



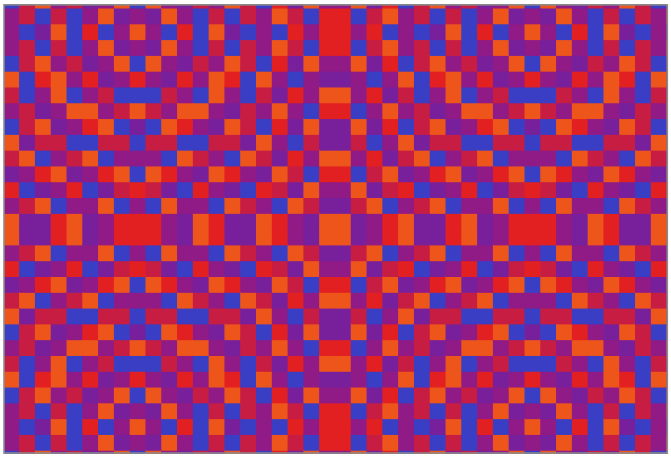
Twill
Colouring: by direction · split complementary (red-orange)

Beadwork

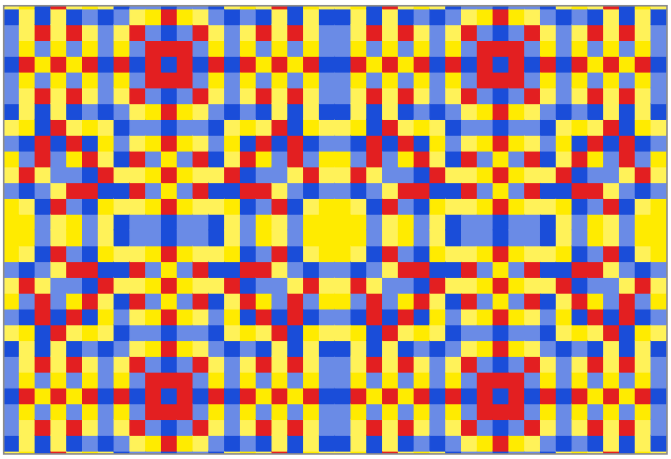
Pixel patterns with the symmetry of the square: a tile of beads whose colour depends on the distance to its centre in a way chosen by a seed (-p seed=...), repeated over the plane as in bead and cross-stitch embroidery.



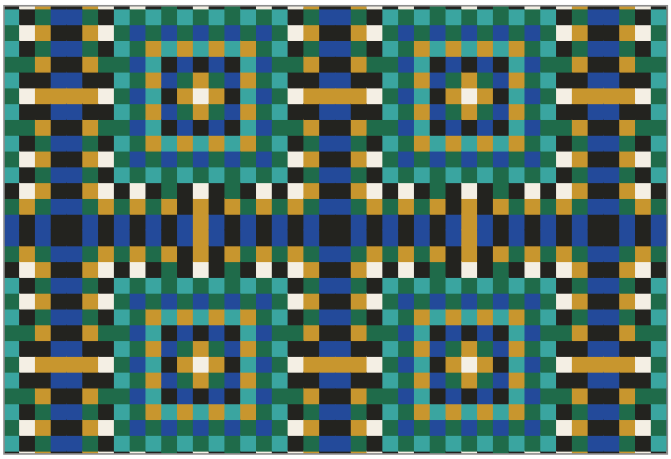
Beadwork, seed 3
Colouring: by distance to the centre · rainbow (yellow)



Beadwork, seed 7
Colouring: by distance to the centre · sweep of hues (red-orange)



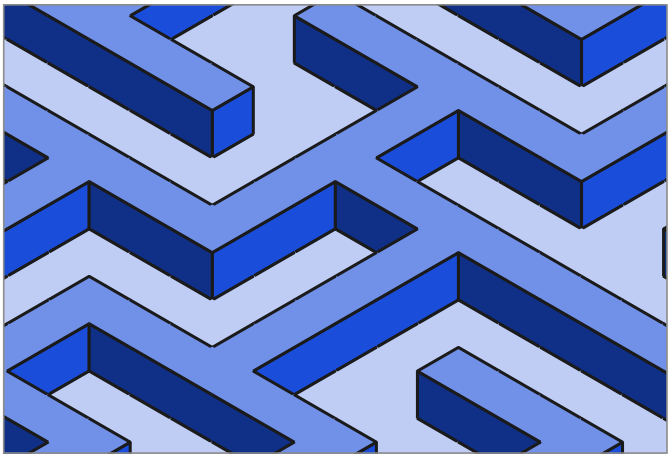
Beadwork, seed 11
Colouring: by distance to the centre · triad (blue)



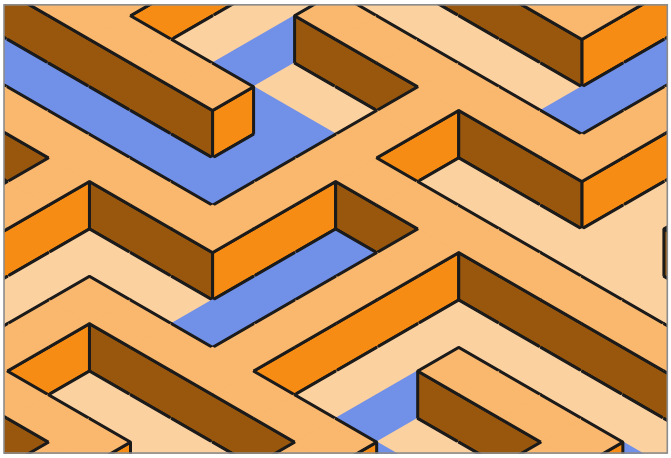
Beadwork in zellij colours
Colouring: by distance to the centre · zellij palette

Mazes

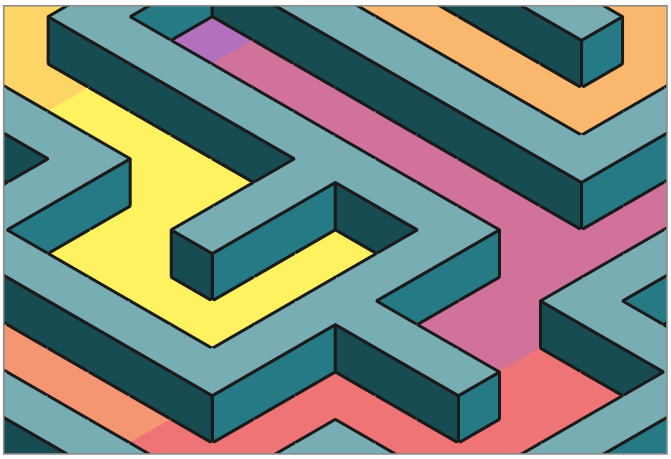
Perfect mazes (exactly one way between any two points) dug at random from a seed (-p seed=...) on a torus, so that the maze repeats without seams. Drawn as walls of cubes in isometric view or flat from above; the floor can follow a ramp of colours along the corridors, or show the longest way through the maze.



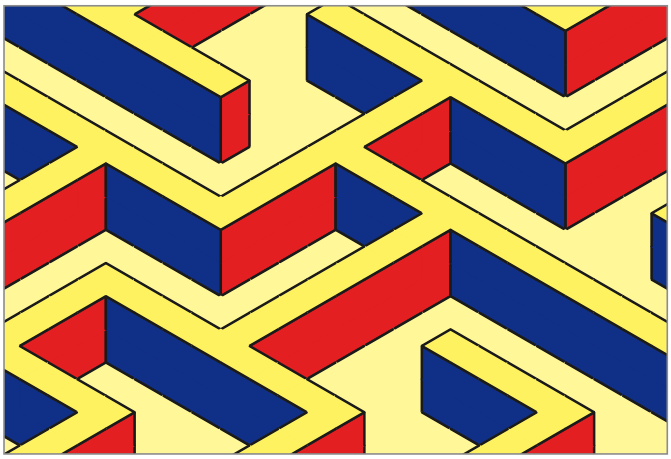
Isometric maze
Colouring: walls and floor · monochromatic (blue)



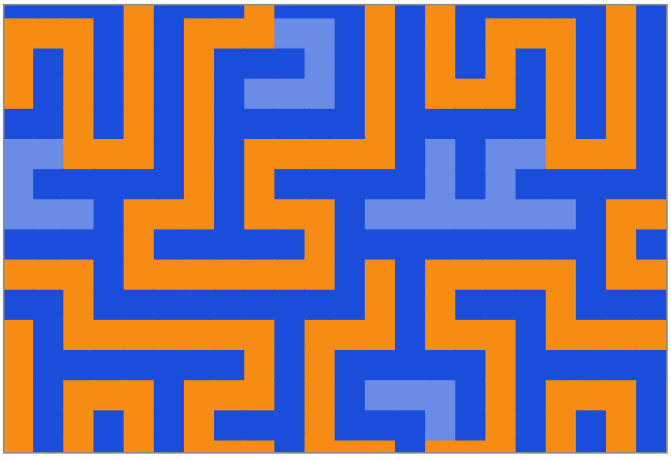
The longest way through the maze
Colouring: longest way · complementary (orange)



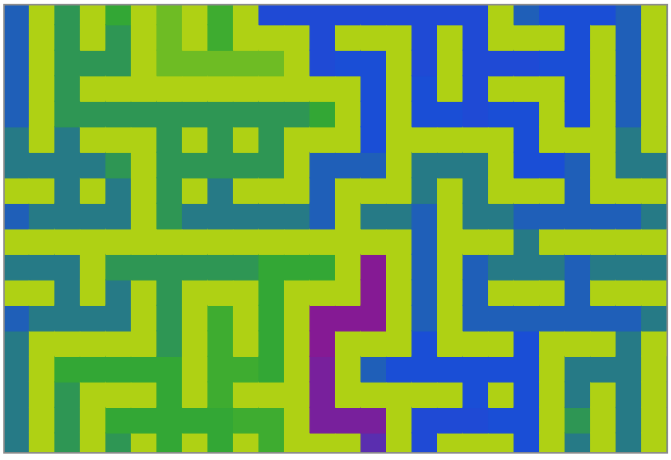
Colours flowing along the corridors
Colouring: along the corridors · rainbow (yellow)



Tall walls, wide corridors
Colouring: by face · triad (yellow)



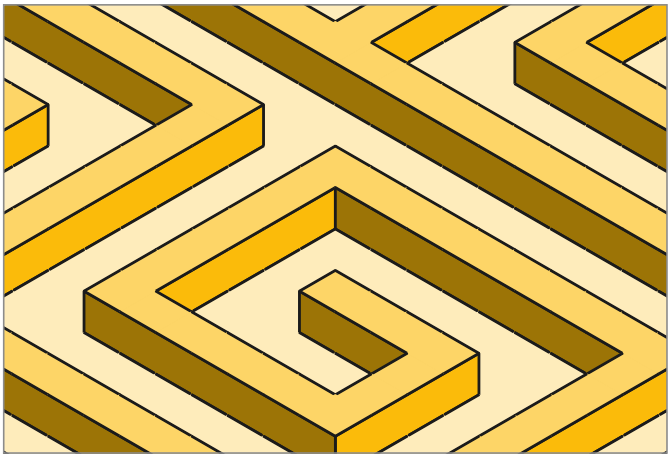
Maze from above, longest way
Colouring: longest way · complementary (blue)



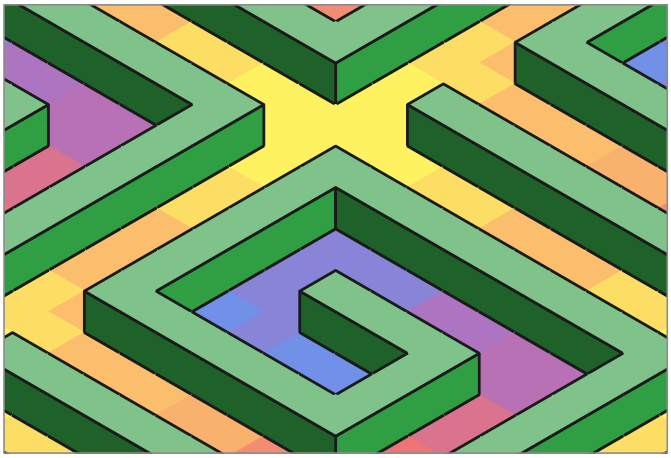
Maze with loops, in a sweep of hues
Colouring: along the corridors · sweep of hues (violet)

Square spirals and meanders

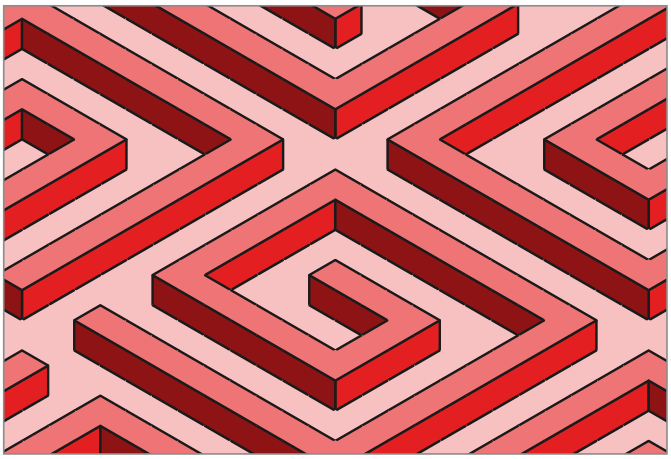
Square spirals of wall, each turn one corridor further in. Joined along the top they make the Greek key (meander); turned a quarter turn each round a corner they make pinwheels of four spirals. The colours can flow along the corridor into each spiral.



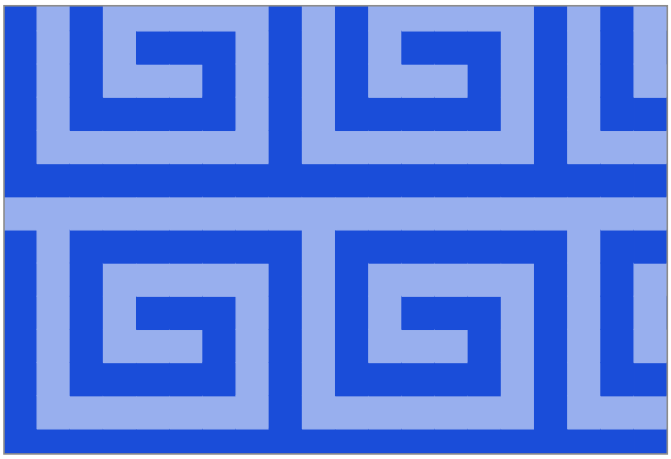
Isometric meander
Colouring: walls and floor · monochromatic (yellow-orange)



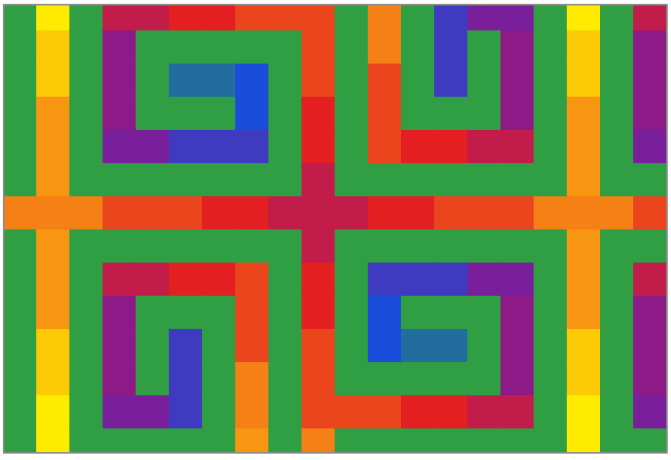
Square spirals, rainbow corridors
Colouring: along the corridors · rainbow (yellow)



Pinwheels of four spirals
Colouring: walls and floor · monochromatic (red)



Greek key
Colouring: walls and floor · monochromatic (blue)



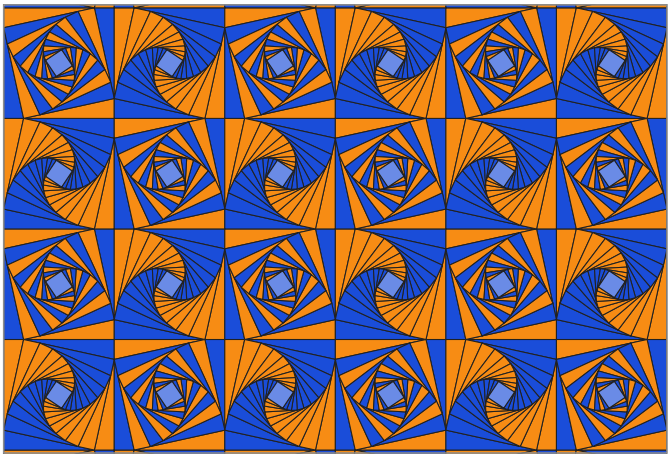
Spiral pinwheels from above
Colouring: along the corridors · rainbow (yellow)



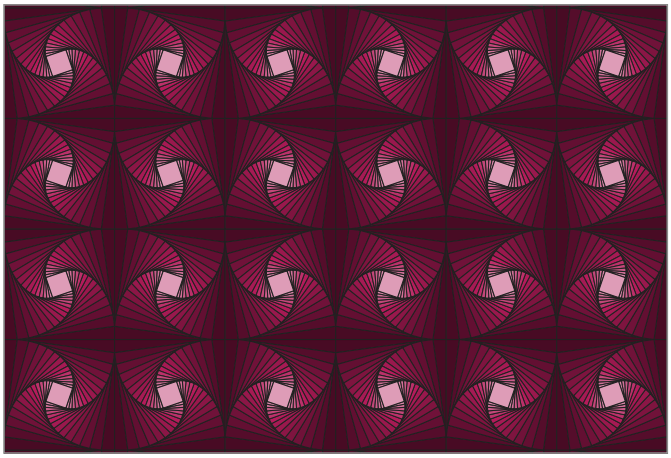
Larger spirals, the longest way
Colouring: longest way · split complementary (blue-green)

Whirling polygons

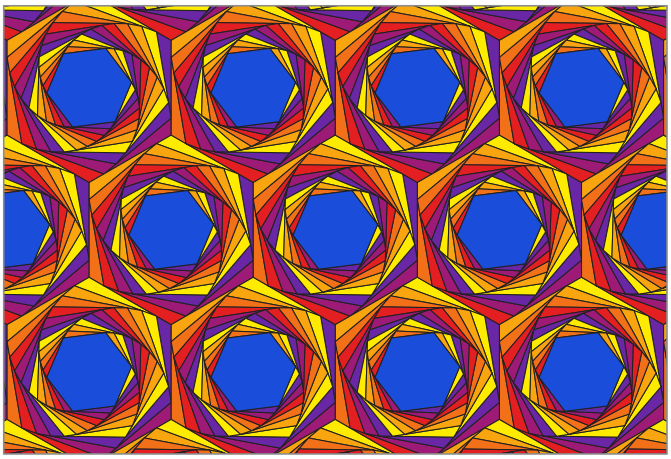
In every tile a polygon whose corners slide a little along its edges, again and again: the cut-off triangles line up in spiral arms. Neighbouring tiles turn the other way, so that the arms run on from tile to tile. Works on any tiling (-p base=...).



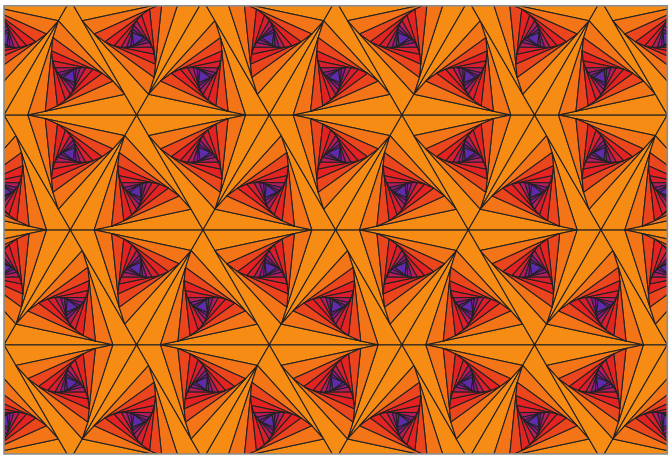
Whirling squares
Colouring: spiral arms · complementary (blue)



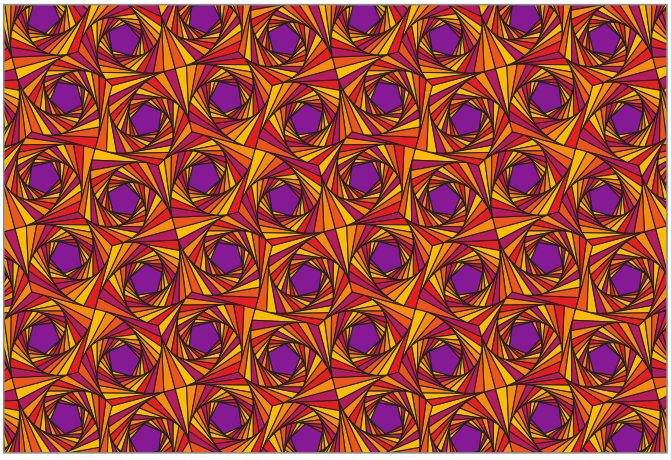
Whirling squares in a ramp
Colouring: by step · ramp, dark to light (red-violet)



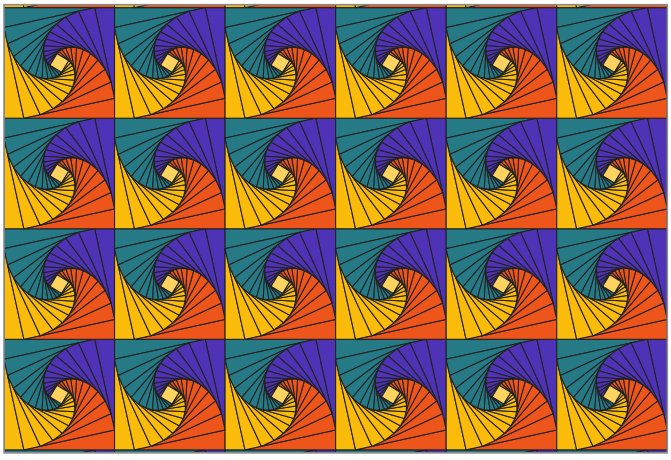
Whirling hexagons
Colouring: spiral arms, other hand · rainbow (yellow)



Whirling triangles
Colouring: by step · sweep of hues (orange)



Whirls on the Cairo tiling
Colouring: spiral arms · analogous (red-orange)



One colour per corner, all turning the same way
Colouring: by corner · tetrad (rectangle) (yellow-orange)